

RETRACE: AGENTIC TEMPORAL-EVIDENCE ARBITRATION FOR CONTEXT-GROUNDED ASR REVISION

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ABSTRACT

Names and technical terms that are acoustically plausible at first mention may become disambiguated only after later dialogue arrives. Revising them with language models, however, risks changing already-correct content, a failure poorly captured by aggregate CER or WER. We present RETRACE, an agentic ASR post-correction system that treats revision as temporal-evidence arbitration. RETRACE preserves immutable first-pass observations, jointly reasons over complete dialogue and entity context, and stores the corrected transcript as an auditable projection. We evaluate RETRACE on a fixed bilingual ContextASR pilot using MOSS-Transcribe-Diarize, Qwen3-ASR-1.7B, and Whisper-large-v3-turbo as first-pass recognizers. Besides CER/WER, we report named-entity false-negative rate and Safe Context Recovery, which balances recovered entity errors against newly introduced ones. Across all six model-language settings, RETRACE reduces transcript error by 3.62–49.66% relative and lowers entity false negatives, achieving SCR-F1 scores of 63.58–94.57%. On MOSS, removing entity context substantially reduces SCR-F1, while a staged composite ablation further limits recovery. These results demonstrate consistent context-grounded gains across three recognizers; English retains a transcript-gain/entity-precision trade-off.

Index Terms— speech recognition, contextual correction, agentic ASR, named entities, safe revision

1. INTRODUCTION

Streaming ASR is often evaluated as though the correct interpretation were available when an utterance ends. In conversation, however, a name or technical term can remain acoustically plausible under several readings until later dialogue supplies discriminative evidence. Figure 1 illustrates the setting: the first-pass ASR output is preserved as an immutable observation, while accumulated dialogue supports an auditable revision of the displayed transcript. Direct language-model rewriting can recover such errors but may also introduce unsupported edits [1, 2, 3].

Prior contextual ASR injects vocabularies, retrieved knowledge, or preceding transcripts to improve the current hypothesis [4, 5, 6, 7]. Memory-augmented correction retrieves earlier history for the current turn [8], while Retrospective Speech Recognition uses subsequent dialogue to refine earlier Japanese ASR output [9]. What remains underexplored is whether a general agent can recover context-dependent entity errors across a complete dialogue without damaging already correct content. This safety question is especially important when the raw error rate is already low: a system may recover a rare name yet still worsen the surrounding transcript.

We introduce RETRACE, an agentic system that treats contextual error correction as temporal evidence arbitration over immutable ASR

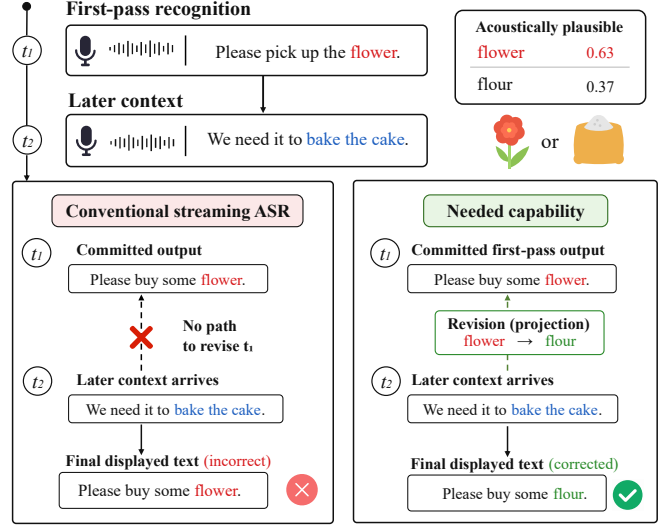


Fig. 1. Motivation for revisable streaming ASR. The first-pass recognizer commits an acoustically plausible but incorrect word at t_1 . A later utterance supplies discriminative context; RETRACE keeps the raw observation immutable and records a separate transcript projection.

observations. As Fig. 2 details, the Agent retrieves dialogue and memory context, identifies conflicts, reasons about their temporal relation, selects available evidence, and records an evidence-grounded revision without overwriting the acoustic observation. These are logical responsibilities of one system, not requirements for separate model calls. In the evaluated implementation, Full RETRACE realizes them jointly through one complete-dialogue Agent call. Its contributions are:

- We formulate contextual ASR correction as agentic temporal-evidence arbitration over immutable first-pass observations, with replayable revision state separated from the acoustic record.
- We instantiate the architecture as Full RETRACE, whose Agent jointly performs context retrieval, conflict analysis, arbitration, and revision over complete dialogue and entity context, and isolate entity evidence through a matched context control.
- We maintain an auditable Raw/revision boundary and evaluate both recovered and newly introduced entity-span errors rather than relying on CER/WER alone.
- We evaluate contextual recovery on Mandarin and English ContextASR dialogue using named-entity false negatives and Safe Context Recovery in addition to conventional CER/WER.

2. RELATED WORK AND PROBLEM SETTING

Contextual ASR and error correction. Contextual biasing improves rare words by injecting or retrieving relevant terms [4, 10]. Retrieval-augmented LLMs extend this to post-correction [5], while generative methods map N -best hypotheses to transcripts [11, 1]. Acoustic fusion constrains plausible but signal-inconsistent edits [12, 2, 13]; Chinese evaluation likewise shows that prompting can introduce errors [3]. These methods primarily correct the current item rather than license a backward edit from later dialogue.

Memory and retrospective evidence. Retrieval-augmented correction improves domain and entity robustness [6, 7], and RAMC-Corr organizes preceding interaction into an ontology memory [8]. RSR uses subsequent dialogue to refine earlier recognition [9]. RETRACE instead exposes an immutable first-pass dialogue and entity context to a global agent, storing its projection separately so recovered and newly introduced entity errors share exactly the same input.

Agentic and stateful ASR. Agentic ASR makes correction an explicit decision process. Interactive ASR uses an LLM-driven multi-turn correction agent and semantic evaluation [14]; Voice Memory lets a frozen corrector act or abstain from auditable domain memory [15]. AgenticASR continually revises a bounded span for disfluency removal, self-correction, and written-form normalization [16]. RETRACE instead performs whole-dialogue post-correction over immutable observations and targets faithful entity recovery: a valid clean-text transformation may be an unsupported edit against a verbatim reference. Thus AASR-Bench is related but not interchangeable with our ContextASR evaluation.

2.1. Problem Formulation

At dialogue step t , a first-pass recognizer emits an immutable observation $o_t = (h_t, \tau_t, m_t)$ containing its hypothesis, time interval, and provenance. Let $O_{1:t}$ denote the accumulated observations and C_t the external contextual information, such as a domain entity list. A correction agent produces a revised projection

$$Y_t = A_\theta(O_{1:t}, C_t, \ell), \quad (1)$$

where ℓ is the language. Raw observations are never overwritten. An accepted revision may be represented by localized, append-only events E_t , from which the displayed transcript is

$$\hat{Y}_t = \text{Replay}(O_{1:t}, E_t). \quad (2)$$

Full RETRACE may jointly realize the same revision semantics and store Y_t as a run-level projection; the staged ablation exposes the localized events explicitly. These are two materializations of the same observation–revision separation, not different problem definitions. The goal is not merely to minimize CER/WER, but to recover context-relevant errors while avoiding newly introduced entity errors and aggregate transcript degradation. In the reported offline setting, t is the end of the recording and $O_{1:t}$ is the complete first-pass dialogue. This supplies both preceding and subsequent textual evidence to the editor, but does not by itself prove which turn caused a correction. No end-to-end recognizer training is assumed.

3. RETRACE: AGENTIC TEMPORAL-EVIDENCE ARBITRATION

3.1. Agentic Arbitration Architecture

Figure 2 decomposes Full RETRACE into logical responsibilities rather than separate model calls. Immutable recognition turns enter

short-term dialogue memory; long-term memory retains stable context and accepted revision state. The Agent retrieves this evidence and interprets an observation as *consistent*, *novel*, *conflicting*, or *uncertain*. It then distinguishes readings that can *coexist*, a genuine *temporal change*, and mutually exclusive alternatives. The first two preserve compatible history, whereas a mutually exclusive or uncertain reading opens evidence verification. This relation step prevents every lexical difference from being treated as an ASR error.

3.2. Immutable First Pass and Context Packet

RETRACE stores recognition turns, timestamps, and provenance once as $O_{1:t}$; model output changes only the derived projection in Eq. (2). Thus fluent correction cannot be mistaken for new acoustic evidence, and every method must consume the same serialized first-pass artifact.

The global context packet is

$$P_t = \{\ell, H_{1:t}, V_t\}, \quad (3)$$

where $H_{1:t}$ is the concatenated first-pass transcript and V_t the deduplicated entity list. Here $H_{1:t}$ supplies dialogue memory for retrospective reasoning and V_t supplies external entity evidence, not insertion commands. The instruction forbids paraphrasing, summarization, translation, alias normalization, and unsupported insertion; reference text and oracle spans are never included.

Full RETRACE jointly realizes retrieval, conflict interpretation, relation reasoning, and revision in one complete-dialogue **Agent** call. Its JSON **text** contains the full transcript; unchanged text is the **keep** decision. Exact comparison derives changed; missing text yields **invalid.output** and returns $H_{1:t}$. This joint call implements Fig. 2.

3.3. Evidence Routes, Resolution, and Replay

For mutually exclusive or uncertain readings, the retrospective route verifies a historical span against later dialogue and entities; the acoustic-sentinel route can request a targeted acoustic check under low confidence or context–audio disagreement. The resolver combines audio support, context support, relation consistency, and confidence to accept, reject, or defer. ContextASR provides complete first-pass dialogue and entities, so our experiments exercise retrospective textual reasoning without additional acoustic re-listening.

Accepted revisions remain separate from observations. A localized ledger event records target span, old/new text, evidence, route, and score; replay constructs the transcript, while rollback changes only the projection. Full RETRACE materializes the same boundary as one run-level record containing the first pass, **Agent** status, and final projection. The ledger is therefore fine-grained in staged execution and coarse-grained in our one-call implementation.

3.4. Staged Composite Ablation

The staged ablation exposes proposal, evidence filtering, and localized commit separately while using the same first pass and entity context. Because all three behaviors change together, it is a composite ablation, not another method or evidence for one gate. Across the figure and experiments, RETRACE remains one method: the figure expands the Agent’s responsibilities and Full RETRACE executes them jointly. Localized source-span and boundary validation is left for future evaluation.



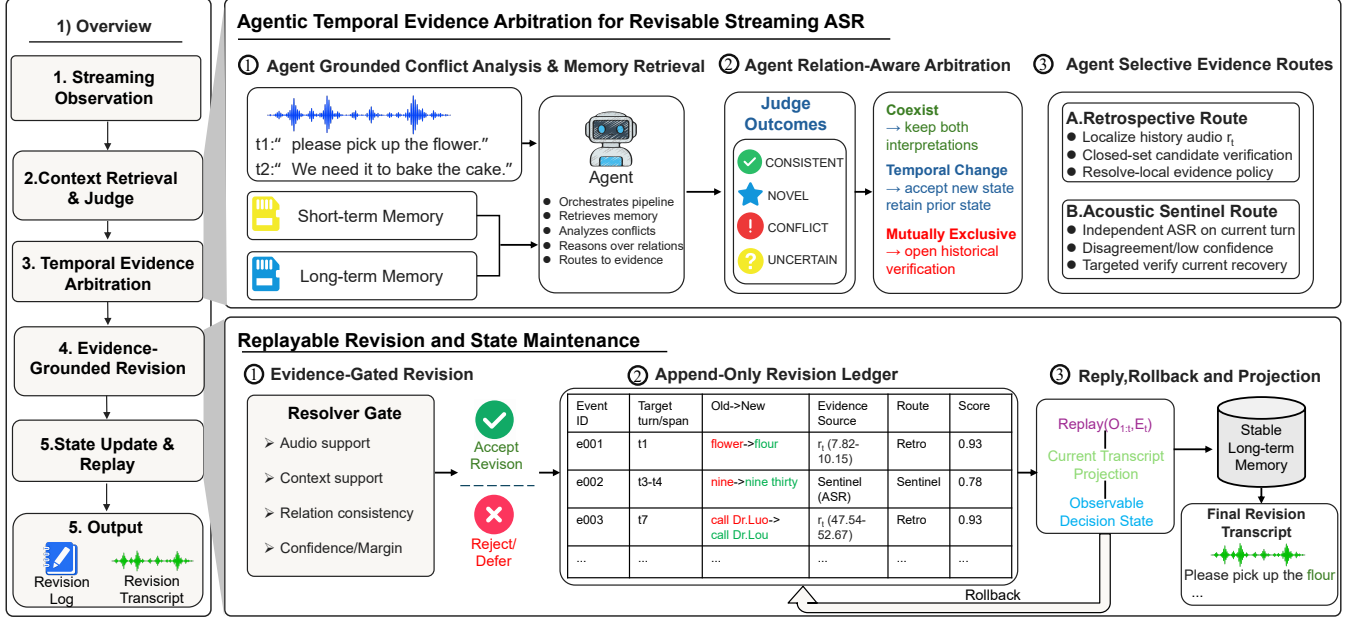


Fig. 2. Full ReTrace architecture. Boxes decompose one Agent system’s logical responsibilities; our implementation realizes them jointly in one complete-dialogue call. ContextASR exercises retrospective textual evidence, while staged execution is only a composite ablation.

4. EXPERIMENTAL SETUP

4.1. Bilingual Contextual-ASR Benchmark

We evaluate on Mandarin and English *ContextASR-Dialogue* from ContextASR-Bench [5], which provides multi-turn audio, reference transcripts, and recording-level fine-grained entity lists. This directly tests whether dialogue and entity context recover errors without damaging correct content. We freeze an evaluation pilot subset and all first-pass outputs before correction, with no added human ambiguity annotation.

References are isolated from inference: the agent receives only language, immutable first-pass text, and the provided entity list. References and annotations are loaded only after outputs are stored. Mandarin uses character normalization and English word tokenization.

4.2. Systems and Controlled Ablation

We evaluate three fixed first-pass front ends: MOSS-Transcribe-Diarize [17], Qwen3-ASR-1.7B [18], and Whisper-large-v3-turbo from the Whisper model family [19]. For each front end, **Raw ASR** returns its frozen transcript unchanged. Full **ReTrace** realizes the architecture in Fig. 2 through one complete-dialogue DeepSeek-Chat Agent call: the Agent receives language, first-pass dialogue, and the fine-grained entity list and jointly performs contextual conflict analysis, arbitration, and revision. The run record retains the immutable first pass, model identity, Agent status, and final projection. No prompt contains the reference transcript, error-span labels, or oracle source-to-target alignments.

On MOSS, two controls use the same frozen first pass. **w/o entity context** keeps Full ReTrace and its complete-dialogue Agent contract unchanged but sends an empty entity list, directly isolating the fine-grained ContextASR hints. **w/o joint Agent revision** keeps the entity context but exposes proposal, evidence filtering, and local-

ized commit as a staged pipeline. The latter is a composite ablation because several coupled responsibilities change together; it is not a separate ReTrace method and is not attributed to one gate. The acoustic-sentinel route is part of the architecture but is not invoked by this text-context experiment, so no acoustic-route gain is claimed.

The external Agent service does not expose deterministic sampling. We execute each correction condition three times, retain all runs, and report metrics from pooled errors, reference units, and entity events. No best-run selection is performed. Within every front end, **Raw** and corrected conditions share manifests, first-pass hashes, text joining, and normalization; a hash mismatch rejects the run.

4.3. Metrics

Conventional transcript quality is measured by character error rate (CER) for Mandarin and word error rate (WER) for English. Because these aggregate rates underweight rare but important names and terms, we additionally report two context-sensitive metric families.

Named-entity false-negative rate (NE-FNR) is the fraction of normalized reference entity occurrences not recovered in the hypothesis. If $M(Y)$ is the number of missed occurrences in transcript Y and N_e the number of reference occurrences, then

$$\text{NE-FNR}(Y) = M(Y)/N_e. \quad (4)$$

Safe Context Recovery (SCR) compares reference-side edit errors inside benchmark entity spans before and after correction. A true positive is a baseline entity error removed by the final transcript; a false positive is a new entity-span error introduced by correction; and a false negative is an original entity error that remains. Pooled over all runs,

$$P_s = \frac{TP}{TP + FP}, \quad R_s = \frac{TP}{TP + FN}, \quad F_s = \frac{2P_s R_s}{P_s + R_s}. \quad (5)$$

Raw ASR has no correction decisions, so SCR precision is undefined and its recovery recall and F_1 are zero by construction.

These metrics separate properties that CER/WER alone conflates: global transcript accuracy, missed reference entities, and the balance between recovered and newly introduced entity-span errors. They are automatic **GT-derived** evaluation measures, not manually adjudicated semantic-ambiguity labels or the official ContextASR scorer. In particular, SCR is deliberately stricter than a clean-text rubric: a formatting or normalization change is not rewarded unless it reduces the reference-aligned entity error.

5. RESULTS AND ANALYSIS

Table 1 shows consistent directional gains across front ends: RETRACE lowers both CER/WER and NE-FNR in all six backbone–language conditions. Mandarin relative CER reduction is 49.66% for MOSS-TD, 31.85% for Qwen3, and 49.53% for Whisper-T; their SCR-F1 values remain closely grouped at 94.57%, 93.07%, and 92.39%. The near-perfect Mandarin SCR precision indicates that the recovered entity errors are rarely offset by new entity-span errors in this pilot.

English exposes a wider trade-off. Relative WER reduction is 22.09% for MOSS-TD, 3.62% for Qwen3, and 23.69% for Whisper-T, while every front end also lowers NE-FNR. Qwen3 English is the weakest global result and attains 63.58% SCR-F1. Whisper-T has the highest English SCR precision (94.29%) but lower recall (51.97%); MOSS-TD recovers more errors (74.14% recall) with lower precision (68.25%). Thus RETRACE transfers across first-pass models, but the balance between recovery coverage and newly introduced entity-span errors depends on the front end.

Table 2 isolates both context and joint Agent reasoning within RETRACE. Removing entity context reduces SCR-F1 from 94.57% to 39.68% in Mandarin and from 71.07% to 47.93% in English, and lowers English relative WER reduction from 22.09% to 11.39%. Thus entity context supports both transcript accuracy and safe entity recovery. Exposing proposal, filtering, and commit as separate stages reduces SCR recall to 15.46%/28.74% and recovers only 45/50 pooled errors, versus 261/129 for Full RETRACE. Full inference also needs one Agent call and lower correction-path latency than staging. These matched controls attribute the main gain to entity context

and joint complete-dialogue arbitration, while the composite staged comparison is not assigned to one internal gate.

5.1. Why Aggregate Error Is Insufficient

The metric gaps reveal where gains occur. Qwen3 English WER changes by only 0.227 points while NE-FNR drops 13.93 points; Whisper-T Mandarin CER drops 4.992 points while NE-FNR drops 59.45 points. Names occupy few tokens, so corpus error understates entity recovery and hides edit composition: MOSS-TD English recovers 129 errors but introduces 60 entity-span errors. CER/WER, NE-FNR, and SCR therefore measure complementary global accuracy, entity misses, and safe recovery.

The practical implication is to retain global context for detection but narrow the commit surface. Output should become localized span edits, rejected when a source is absent or non-unique, crosses turns, reorders broadly, or normalizes unsupported forms. This untested safety layer follows from the error audit and requires a frozen ablation.

5.2. Limitations

This fixed bilingual pilot is not full-benchmark evaluation. SCR and NE-FNR are exact-match GT measures rather than semantic judgments. The staged control jointly changes several Agent responsibilities and therefore cannot attribute its difference to any single internal operation. Repeated Agent runs measure service variability rather than independent samples. **The evidence supports cross-front-end contextual correction and the two controlled factors, without making a state-of-the-art claim.**

6. CONCLUSION

RETRACE is an agentic temporal-evidence arbitration method for auditable ASR revision. On bilingual ContextASR, its joint complete-dialogue implementation improves transcript and entity accuracy across three recognizers. Matched controls show reliance on entity hints and joint reasoning. English SCR reveals a WER-gain/entity-precision trade-off, motivating localized validation and acoustic-route evaluation.

Table 1. Multi-backbone ContextASR results. Error is CER for Mandarin (Zh) and WER for English (En); arrows show Raw→RETRACE. Corrected metrics pool errors, reference units, and entity events over three runs.

Front end	Lang.	Error ↓	NE-FNR ↓	SCR-P	SCR-R	SCR-F1
MOSS-TD	Zh	1.763→ 0.888	28.72→ 3.55	100.00	89.69	94.57
Qwen3	Zh	2.992→ 2.039	30.50→ 3.19	100.00	87.04	93.07
Whisper-T	Zh	10.080→ 5.088	65.60→ 6.15	99.86	85.96	92.39
MOSS-TD	En	2.431→ 1.894	19.92→ 6.90	68.25	74.14	71.07
Qwen3	En	6.251→ 6.024	26.95→ 13.02	73.33	56.12	63.58
Whisper-T	En	7.776→ 5.934	35.94→ 15.63	94.29	51.97	67.01

Table 2. MOSS-TD context, pipeline, and efficiency controls. **Rel. red.** is computed from unrounded pooled error rates against the shared Raw ASR input; TP/FP/FN pool three runs. Calls are Agent-service calls and Time is correction-path latency (seconds), both averaged per sample-run and excluding frozen first-pass ASR. Staging jointly changes proposal, filtering, and commit and is therefore a composite ablation.

Lang.	Variant	Transcript quality		Safe recovery				Efficiency		
		Raw→Output	Rel. red.	SCR-P	SCR-R	SCR-F1	TP	FP	FN	Time
Zh	Full RETRACE	1.763→ 0.888	49.66%	100.00	89.69	94.57	261	0	30	2.67
Zh	w/o entity context	1.763→1.490	15.49%	86.21	25.77	39.68	75	12	216	2.52
Zh	Staged ablation	1.763→1.570	10.94%	100.00	15.46	26.79	45	0	246	4.28
En	Full RETRACE	2.431→ 1.894	22.09%	68.25	74.14	71.07	129	60	45	2.65
En	w/o entity context	2.431→2.154	11.39%	49.39	46.55	47.93	81	83	93	1.00
En	Staged ablation	2.431→2.204	9.32%	59.52	28.74	38.76	50	34	124	6.32

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